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Data Analytics with R

Quality Laboratory Manual

Experiment No. 04

To implement a program to read data from a CSV file and clean it using R



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Experiment No. 04

Title of Experiment: To implement a program to read data from a CSV file and perform data cleaning using R (Handling missing values, removing duplicates and renaming columns)

Aim of Experiment: To read data from a CSV file into R and perform essential data cleaning operations such as handling missing values, removing duplicate records and renaming columns to prepare the dataset for analysis

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Internet Connection (for dataset access)/ CSV dataset file

Experiment Objectives:

- To import datasets from CSV files into R
- To inspect dataset structure and summary statistics
- To identify missing values in datasets
- To apply techniques to handle missing data
- To detect and remove duplicate records
- To rename columns for better readability
- To prepare cleaned datasets suitable for further analysis

Experiment Outcomes:

After completing the experiment, students will be able to –

- Load external datasets into R environment
- Perform data preprocessing and cleaning tasks
- Handle incomplete and inconsistent data
- Improve dataset quality before analysis
- Understand the importance of data cleaning in analytics workflows

Theory:**1. Data Import in R**

Real-world datasets are commonly stored in CSV (Comma-Separated Values) format. R provides functions to import such datasets into data frames.

Example:

```
data <- read.csv("data.csv")
```

A **data frame** is the primary structure used for storing tabular data in R.

2. Data Cleaning

Data cleaning is the process of identifying and correcting errors, inconsistencies, and missing values in datasets. It is a critical step in data analytics because poor data quality leads to incorrect analytical results.

3. Handling Missing Values

Missing values in R are represented using `NA`.

Common methods:

- Detect missing values:
`is.na(data)`
- Count missing values:
`sum(is.na(data))`
- Remove missing records:
`na.omit(data)`
- Replace missing values:
`data$Age[is.na(data$Age)] <- mean(data$Age, na.rm = TRUE)`

4. Removing Duplicate Records

Duplicate data can distort analysis results.

```
data <- unique(data)
```

or

```
data <- data[!duplicated(data), ]
```

5. Renaming Columns

Column names can be modified for clarity.

```
colnames(data)
```

```
colnames(data)[1] <- "Student_ID"
```

6. Data Inspection Functions

Important exploratory functions:

Function	Purpose
<code>head()</code>	View first rows
<code>str()</code>	Structure of dataset
<code>summary()</code>	Statistical summary
<code>dim()</code>	Dataset dimensions

Task to Perform**Case Study-Based Observation Task – Cleaning Customer Sales Data for Business Analytics**

A retail company collects customer sales data from multiple store branches across different cities. Due to manual data entry and system integration issues, the dataset contains several inconsistencies such as missing values, duplicate records, and unclear column names.

Before performing sales analysis and visualization, the data analyst must clean and prepare the dataset using R to ensure accuracy and reliability.

You are assigned as a Data Analyst to preprocess the dataset using data cleaning techniques in R.

Dataset Description (Sales_Data.csv)

The dataset contains the following fields:

Column Name	Description
Cust_ID	Customer identification number
CustName	Customer name
City	Customer city
Purchase_Amt	Purchase amount
Age	Customer age
Payment_Mode	Payment method
Purchase_Date	Date of purchase

Task-1 –**Load Dataset**

Import CSV file into R

Observe the dataset structure using – str() and head()

Task-2 –

Identify Missing Values – Detect missing values and count total missing entries and record the observations

Task-3 –**Handle Missing Values**

Perform appropriate changes –

- Replace missing Age values with mean age
- Replace missing Purchase_Amt with median purchase value

Task-4 –**Remove Duplicate Records**

Observe and note the dataset size before and after removal

Task-5 –**Rename Columns**

Provide a standardized column names

Task-6 –**Verify Cleaned Dataset**

Check final dataset using summary() and dim() functions

Submission Guidelines

- 1) Include R code and outputs
- 2) Submit report (2-3 pages PDF)

Observations:

- CSV data was successfully imported into R as a dataframe
- Missing values were detected using is.na() function
- Replacing missing values improved dataset completeness
- Duplicate entries were identified and removed
- Column renaming improved dataset readability
- Cleaned data became suitable for further analysis and visualization

Conclusion:

The experiment demonstrated how to import and clean datasets using R. students learned to identify missing values, remove duplicates and rename columns effectively. Data cleaning ensures improved data quality and reliability, forming a crucial preprocessing step before performing analytical or predictive tasks

Expected Oral Questions:

- 1) What is a CSV file?
- 2) Which function is used to read CSV file data in R?
- 3) How are missing values represented in R?
- 4) What is data cleaning?
- 5) Why should duplicate records be removed?
- 6) Difference between NA and NULL in R?
- 7) What is a data frame?
- 8) Why is preprocessing necessary before analysis?

FAQs in Interview:

- 1) Why is data cleaning important in data analytics?
- 2) How do you detect missing values in R?
- 3) How can missing values be handled?
- 4) What problems do duplicate records cause?
- 5) Which R structure stores CSV data after import?